WORKED SOLUTION

STEP III 2025, Question 2

Pure • Iterated Functions and Periodicity

STEP III 2025, Question 2

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2025 STEP papers (a zip file):

www.ocr.org.uk/Images/750486-2025-past-papers.zip

This question has a full guided treatment in *Beyond STEP*, Section 8.8, *Period Three Implies Chaos*: https://beyondstep.keithmcnulty.org/ch08/period-three.html

The key idea

Every function in this question is built from straight lines, so the whole question is about keeping track of which piece you are on. Work out $\mathrm{f}(\mathrm{f}(x))$ and $\mathrm{f}(\mathrm{f}(\mathrm{f}(x)))$ piece by piece, using the sign of the inner function to remove each modulus. Then read the sequence questions backwards: a solution of $\mathrm{f}^n(x) = x$ is a starting value that comes back to itself after n steps, and every number less than 7 has exactly two preimages under f, which are negatives of each other.

Solution

Throughout, write $\mathrm{f}^2(x) = \mathrm{f}(\mathrm{f}(x))$ and $\mathrm{f}^3(x) = \mathrm{f}(\mathrm{f}(\mathrm{f}(x)))$. All three functions are even, because f depends only on $|x|$, so it is enough to work for $x \geq 0$ and reflect in the y-axis.

Part (i)

(a) The graph of $y = \mathrm{f}(x)$ is an inverted V with vertex $(0, 7)$, gradients ∓ 2, crossing the x-axis at $x = \pm\frac{7}{2}$.

For f^2 we need the sign of $\mathrm{f}(x)$. If $0 \leq x \leq \frac{7}{2}$ then $\mathrm{f}(x) = 7 - 2x \geq 0$, so

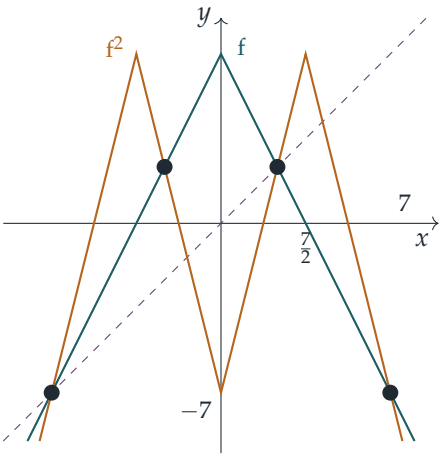
$$\mathrm{f}^2(x) = 7 - 2(7 - 2x) = 4x - 7.$$

If $x \geq \frac{7}{2}$ then $\mathrm{f}(x) \leq 0$, so $|\mathrm{f}(x)| = 2x - 7$ and

$$\mathrm{f}^2(x) = 7 - 2(2x - 7) = 21 - 4x.$$

So $y = \mathrm{f}^2(x)$ is a W shape: value -7 at $x = 0$, peaks of height 7 at $x = \pm\frac{7}{2}$ (exactly where f vanishes), gradients ± 4, and x-intercepts at $\pm\frac{7}{4}$ and $\pm\frac{21}{4}$. Solving $7 - 2x = 4x - 7$ and

$7 - 2x = 21 - 4x$ shows that the two graphs cross at $(\pm\frac{7}{3}, \frac{7}{3})$ and $(\pm 7, -7)$.



(Teal: $y = \mathrm{f}(x)$; amber: $y = \mathrm{f}^2(x)$; dashed: $y = x$.)

(b) We solve on each of the four pieces and keep only roots in the right range.

- $0 \le x \le \frac{7}{2}$: $4x - 7 = x$ gives $x = \frac{7}{3}$, which is in range.
- $x \ge \frac{7}{2}$: $21 - 4x = x$ gives $x = \frac{21}{5} = 4.2$, in range.
- $-\frac{7}{2} \le x \le 0$: here $\mathrm{f}^2(x) = -4x - 7$, and $-4x - 7 = x$ gives $x = -\frac{7}{5}$, in range.
- $x \le -\frac{7}{2}$: here $\mathrm{f}^2(x) = 21 + 4x$, and $21 + 4x = x$ gives $x = -7$, in range.

So $\mathrm{f}(\mathrm{f}(x)) = x$ has exactly four solutions:

$$x = -7, \quad -\tfrac{7}{5}, \quad \tfrac{7}{3}, \quad \tfrac{21}{5}.$$

(c) The sequence has period 2 exactly when $u_2 = u_0$, that is $\mathrm{f}^2(a) = a$, but $u_1 \ne u_0$, that is $\mathrm{f}(a) \ne a$.

Ask yourself Do all four answers to (b) give sequences of period 2?

No. Two of them are fixed points of f itself: $\mathrm{f}(\frac{7}{3}) = 7 - \frac{14}{3} = \frac{7}{3}$ and $\mathrm{f}(-7) = 7 - 14 = -7$. These give constant sequences, which have period 1. The other two swap:

$$\mathrm{f}\left(-\tfrac{7}{5}\right) = 7 - \tfrac{14}{5} = \tfrac{21}{5}, \qquad \mathrm{f}\left(\tfrac{21}{5}\right) = 7 - \tfrac{42}{5} = -\tfrac{7}{5}.$$

So the sequence has period 2 exactly when $a = -\frac{7}{5}$ or $a = \frac{21}{5}$.

(d) With $a = \frac{28}{5}$,

$$u_1 = 7 - \tfrac{56}{5} = -\tfrac{21}{5}, \qquad u_2 = 7 - \tfrac{42}{5} = -\tfrac{7}{5}, \qquad u_3 = \mathrm{f}\left(-\tfrac{7}{5}\right) = \tfrac{21}{5}.$$

From u_2 on, the sequence is the sequence of part (c) starting at $-\frac{7}{5}$, so $u_2, u_3, u_4, \ldots$ is $-\frac{7}{5}, \frac{21}{5}, -\frac{7}{5}, \ldots$ and has period 2. Finally $u_0 = \frac{28}{5}$ and $u_1 = -\frac{21}{5}$ are both different from

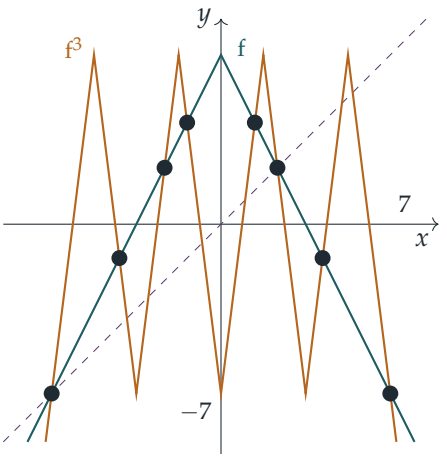
each of $u_2 = -\frac{7}{5}$ and $u_3 = \frac{21}{5}$. (We have to check all four pairs, not just $u_0 \neq u_2$ and $u_1 \neq u_3$.) This is what was required.

Part (ii)

(a) Now apply f to $\mathrm{f}^2(x)$, splitting each piece of f^2 where it changes sign, at $x = \frac{7}{4}$ and $x = \frac{21}{4}$. For $x \geq 0$:

$$\mathrm{f}^3(x) = 7 - 2\left|\mathrm{f}^2(x)\right| = \begin{cases} 7 + 2(4x - 7) = 8x - 7, & 0 \leq x \leq \frac{7}{4}, \\ 7 - 2(4x - 7) = 21 - 8x, & \frac{7}{4} \leq x \leq \frac{7}{2}, \\ 7 - 2(21 - 4x) = 8x - 35, & \frac{7}{2} \leq x \leq \frac{21}{4}, \\ 7 + 2(21 - 4x) = 49 - 8x, & x \geq \frac{21}{4}. \end{cases}$$

So the graph zig-zags between -7 and 7 with gradients ± 8: troughs at $(0, -7)$ and $(\pm\frac{7}{2}, -7)$, peaks at $(\pm\frac{7}{4}, 7)$ and $(\pm\frac{21}{4}, 7)$, then falls away for $|x| > \frac{21}{4}$. Solving $7 - 2x$ against each piece, it crosses $y = \mathrm{f}(x)$ at $(\pm\frac{7}{5}, \frac{21}{5})$, $(\pm\frac{7}{3}, \frac{7}{3})$, $(\pm\frac{21}{5}, -\frac{7}{5})$ and $(\pm 7, -7)$.



(Teal: $y = \mathrm{f}(x)$; amber: $y = \mathrm{f}^3(x)$; dashed: $y = x$.)

(b) Starting from $a = 1$: $u_1 = 7 - 2 = 5$, $u_2 = 7 - 10 = -3$, $u_3 = 7 - 6 = 1$. Starting from $a = -\frac{7}{9}$:

$$u_1 = 7 - \tfrac{14}{9} = \tfrac{49}{9}, \qquad u_2 = 7 - \tfrac{98}{9} = -\tfrac{35}{9}, \qquad u_3 = 7 - \tfrac{70}{9} = -\tfrac{7}{9}.$$

So both sequences return to their start after three steps: they are the 3-cycles $1 \to 5 \to -3 \to 1$ and $-\frac{7}{9} \to \frac{49}{9} \to -\frac{35}{9} \to -\frac{7}{9}$.

Every term of a 3-cycle satisfies $\mathrm{f}^3(x) = x$, not just the starting value, so this gives six solutions 1, 5, -3, $-\frac{7}{9}$, $\frac{49}{9}$, $-\frac{35}{9}$. Also any fixed point of f satisfies $\mathrm{f}^3(x) = x$, which adds $\frac{7}{3}$ and -7 from part (i). That makes eight distinct solutions.

Ask yourself How do we know there are no more?

The formula in (ii)(a) shows that f^3 is made of eight straight pieces (four for $x \geq 0$ and their mirror images), each with gradient ± 8. Since no piece is parallel to $y = x$, each piece meets the line $y = x$ at most once, so there are at most eight solutions. We have found eight, so the complete solution set of $f(f(f(x))) = x$ is

$$x = -7, \quad -\tfrac{35}{9}, \quad -3, \quad -\tfrac{7}{9}, \quad 1, \quad \tfrac{7}{3}, \quad 5, \quad \tfrac{49}{9}.$$

(The sketch agrees: the dashed line crosses each of the eight amber segments once.)

(c) We copy the idea of (i)(d) and work backwards from a 3-cycle. Since $f(y) = c$ means $|y| = \frac{1}{2}(7 - c)$, every $c < 7$ has exactly two preimages, $\pm\frac{1}{2}(7 - c)$. Take $u_3 = 1$ from the cycle $1 \to 5 \to -3 \to 1$. The preimage of 1 inside the cycle is -3, so to leave the cycle we choose the other preimage, $u_2 = 3$, which is not in the cycle. Then

$$f(u_1) = 3 \;\Rightarrow\; |u_1| = 2, \quad \text{take } u_1 = 2; \qquad f(u_0) = 2 \;\Rightarrow\; |u_0| = \tfrac{5}{2}, \quad \text{take } u_0 = \tfrac{5}{2}.$$

So $a = \frac{5}{2}$ works. Checking forwards:

$$\tfrac{5}{2},\ 2,\ 3,\ 1,\ 5,\ -3,\ 1,\ 5,\ -3,\ \ldots$$

From u_3 on, the sequence runs round the cycle $1, 5, -3$ and has period 3, while $u_0, u_1, u_2 = \frac{5}{2}, 2, 3$ are all different from 1, 5 and -3.

(Other choices also work: for example $a = -\frac{5}{2}$ or $a = \pm\frac{9}{2}$, which lead to $u_1 = -2$, or $a = \frac{3}{2}$, which enters the cycle at $u_3 = 5$ via $u_2 = -1$.)

Checks

- Each claimed solution can be iterated by hand: for instance $\frac{49}{9} \to -\frac{35}{9} \to -\frac{7}{9} \to \frac{49}{9}$, and $\frac{28}{5} \to -\frac{21}{5} \to -\frac{7}{5} \to \frac{21}{5} \to -\frac{7}{5}$.
- A computer scan of $f^3(x) - x$ over $-30 \leq x \leq 30$ finds sign changes only near -7, -3.889, -3, -0.778, 1, 2.333, 5 and 5.444, matching the eight solutions; the same scan for $f^2(x) - x$ finds only -7, -1.4, 2.333 and 4.2.
- Counting check: f^n has 2^n linear pieces of gradient $\pm 2^n$, so $f^n(x) = x$ has 2^n solutions. For $n = 2$ that is $4 = 2 + 2$ (two fixed points and one 2-cycle); for $n = 3$ it is $8 = 2 + 3 + 3$ (two fixed points and two 3-cycles).

Common slips

- Giving all four solutions of (i)(b) in (i)(c): the fixed points give period 1.
- In (ii)(b), listing only 1 and $-\frac{7}{9}$, or forgetting that the fixed points $\frac{7}{3}$ and -7 also satisfy $f^3(x) = x$.
- Misplacing where the graphs cross in the sketches; finding the crossing points algebraically fixes this.

Beyond STEP

This is a first taste of discrete dynamical systems. Since f is a tent map with slope 2, f^n has 2^n pieces, so there are periodic points of every period, and the map is a standard example of chaotic behaviour. Sharkovskii's theorem says that for a continuous map of the line, having a point of period 3 forces points of every period, the famous slogan "period three implies chaos". Strogatz, *Nonlinear Dynamics and Chaos*, gives a readable introduction to iterated maps.

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